

17-06-2026

Agenda:

ML-8

- Bias & variance tradeoff
- logistic regression

Bias & variance tradeoff

↓ ↓
ML model / formula / fit

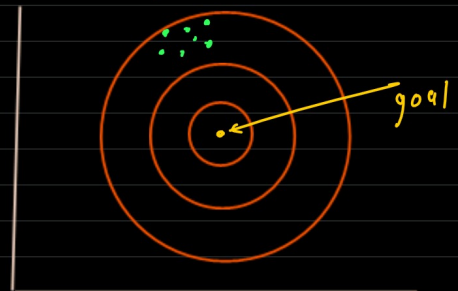
→ use to describe model performance

Bias:
underfitting

train-loss ≈ 0

train-loss $\approx > 0$ poor performance

biased
towards
top left
corner



Not ideal:

→ high bias model

Ideal case:

→ low bias model

variance: -

overfitting

overreacting to the noise

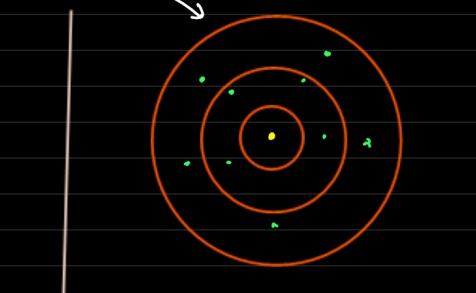
Not ideal:

high variance

Ideal:

low variance

noise → outliers

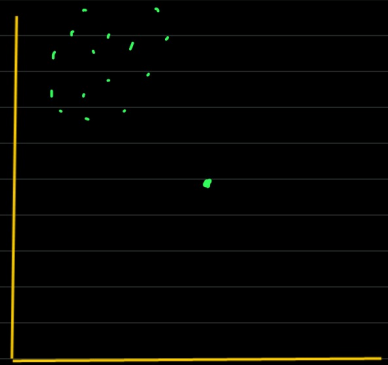


4 stages

High bias, High variance

(worst case): Model is wrong & unstable

Train: High Test: Higher

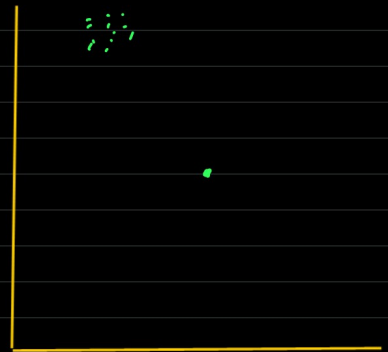


High bias, low variance

(underfitting): model is too simple & misses the truth completely but does so consistently

Train: High

Test: High



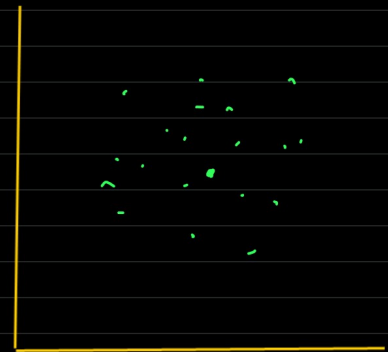
Low bias, High variance

(overfitting) (Bad)

: The model memorized the training data & fails on new data.

Train: Near zero

Test (on): very high

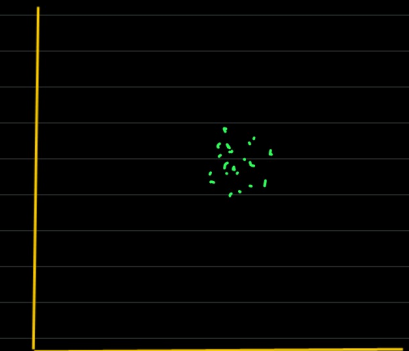
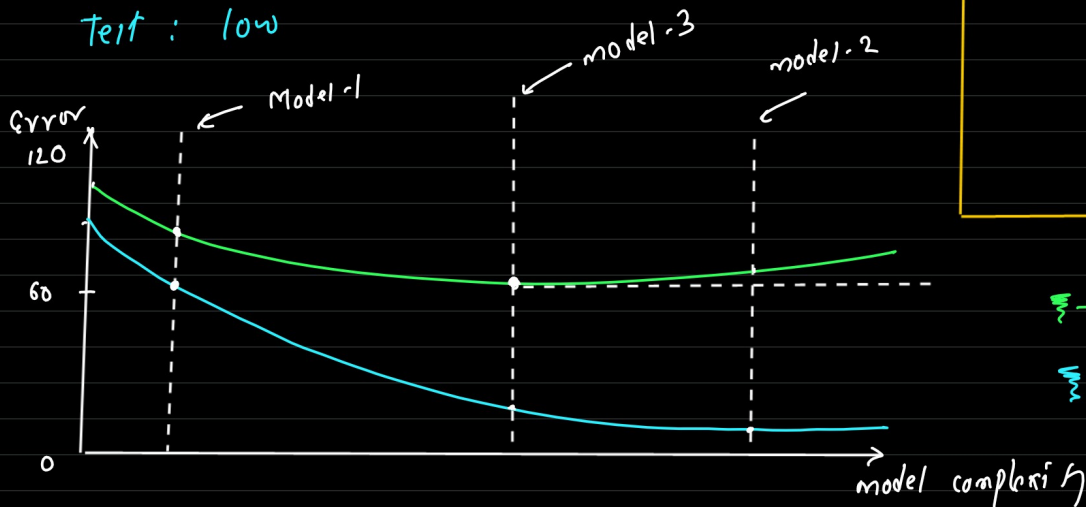


Low bias & low variance (Ideal)

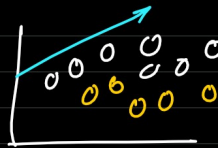
: The model learned the true pattern & ignores the noise

Train: low

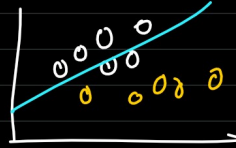
Test: low



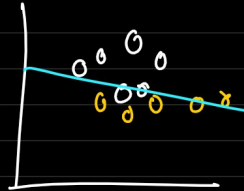
Model-1 $\rightarrow$ underfitted



Model-2 $\rightarrow$ overfitted



Model-3 $\rightarrow$ optimal/ideal



Bias $\rightarrow$ fit of train data
variance $\rightarrow$ Test performance

low bias $\rightarrow$ good fit on train

low variance $\rightarrow$ good fit on test

Train : 80 $\rightarrow$ ML-1
Test : 20

High bias
low variance

Train : 30 $\rightarrow$ low bias
Test : 100 $\rightarrow$ high variance

ML-2
Train : 400
Test : 400

ML-3
Train : 30
Test : 35

Train : 300 $\rightarrow$ High bias
Test : 100 $\rightarrow$ low variance

High bias
High variance

T	100	300
T	400	600



Machine learning algorithm - 2

logistic regression → classification of binary class

0, 1
True / False

activation
fn

$$y = mx + c$$

logistic regression → linear regression + activation function

linear regression

marks in %
y = 80
81
82
83
90

$$\hat{y} \rightarrow \text{mse} \rightarrow (y - \hat{y})^2$$

loss → error

derivative

gradient → direction + magnitude

logistic regression

sick → 1
not sick → 0

$x_1 \quad x_2 \quad x_3 \quad x_4$

y

$$y = m_1 x_1 + m_2 x_2$$

1

0

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

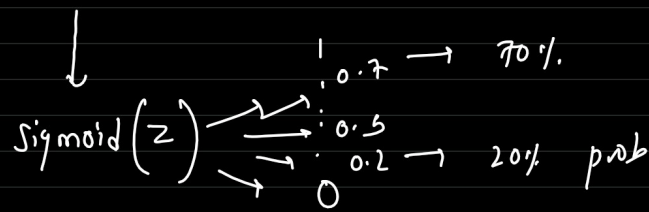
1

→ $y \quad \hat{y}$
1 0
0 1

$$(y - \hat{y})^2 \rightarrow (1)^2$$

$$(y - \hat{y})^2 \rightarrow (1)^2$$

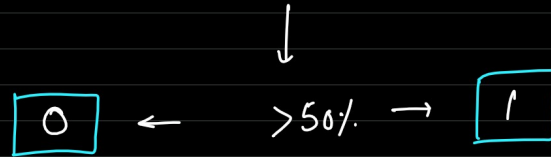
linear regression $\leftarrow z = mx + c$



sick

$>60 \rightarrow \text{sick} \rightarrow 1$

$>80 \rightarrow \text{sick}$



$$J(\beta) = -\frac{1}{N} \sum_{i=1}^N \left(y_i \cdot \log(\hat{y}_i) + (1-y_i) \cdot \log(1-\hat{y}_i) \right)$$

Input

$x_1 \quad y \rightarrow m=1 \rightarrow z = mx_1 \rightarrow \text{sigmoid}(z) \rightarrow 0.9 \leftarrow \underline{\underline{y_{pred}}}$
 $\boxed{0} \quad 0$
 $\rightarrow \text{Actual}$
 $y=0, \hat{y}=0.9$
 $y=1$

$$- y_i \cdot \log(\hat{y}_i) + (1-y_i) \cdot \log(1-\hat{y}_i)$$

$$\rightarrow - [0 \cdot \log(0.9) + (1-0) \cdot \log(0.1)]$$

$$\rightarrow - [0 + 1 \cdot \log(0.1)]$$

$$\rightarrow - \log(0.1)$$

$$\rightarrow - (-2.3025) \rightarrow 2.3025$$

$$\text{MSE} = (y - \hat{y})^2 = 0.81$$

$$y = 0, \hat{y}_{\text{-pred}} = 0.1$$

$$= y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)$$

$$= [0 \cdot \log(0.1) + (1 - 0) \cdot \log(1 - 0.1)]$$

$$= [0 + 1 \cdot \log(0.9)]$$

$$= \log(0.9) \rightarrow -(-0.10536) \rightarrow 0.10536$$

$$\text{MSE} = (y - \hat{y})^2$$

$$= \underline{\underline{0.01}}$$

		new log	MSE
case 1	$(y=0, y_{\text{-pred}}=0.9)$	2.3025	0.81
case 2	$(y=0, y_{\text{-pred}}=0.1)$	0.10536	0.01

flow of logistic regression:

- 1) $x_1 \ x_2 \ x_3 \ x_4 \rightarrow$ linear equation for intermediate o/p $\rightarrow z$
 $\rightarrow m_1 \ m_2 \ m_3 \ m_4$
- 2) $z \rightarrow$ sigmoid $\rightarrow \hat{y}$
- 3) loss-function $(y, \hat{y}) \rightarrow$ error
- 4) loss-function gradient $(y, \hat{y}) \rightarrow$ direction & magnitude
 $\downarrow$
 learning rate $\rightarrow$ update parameters

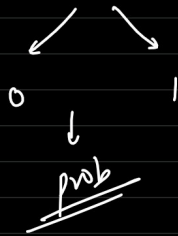
(5) calculate final log using log/loss function

(6) Iterate (1)(2)(4) until condition is satisfied

(7) Done

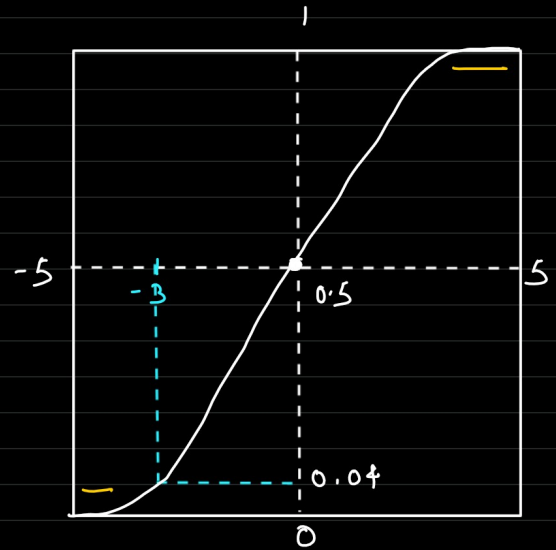
Sigmoid : —

$-\infty$ to $+\infty$

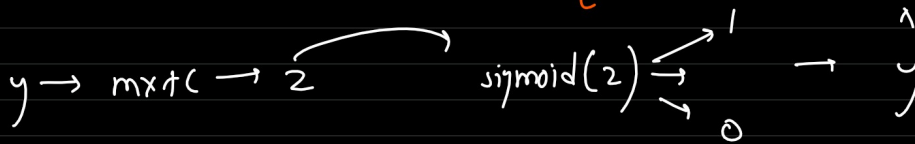


$$\rightarrow \frac{1}{1 + e^{-z}}$$

$$z = x(m) + c$$



activation function



y	$\hat{y}$	$\hat{y} > 0.5$
1	0.9	1
0	0.2	0
1	0.8	1
0	0.3	0